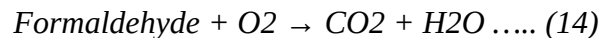
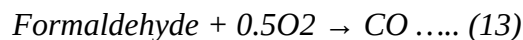
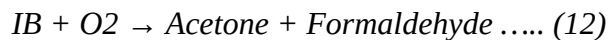
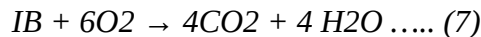
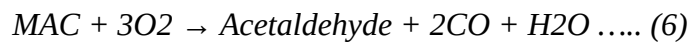
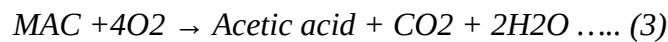
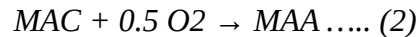


Chapter 8: Equipment Sizing

For the equipments given in Table 18, the complete mechanical design was done for Reactor R201 and the extraction column T204.

8.1 Reactor R201

The feed to the reactor is stream 4, containing Isobutene (IB), steam and air entering at 350 °C. The reactor is a multi-tubular packed bed reactor, where the catalyst used is a MoVBiCsRh oxide. The outlet contains unreacted Isobutene, Methacrolein (MAC), methacrylic acid (MAA), acetone, formaldehyde, acetaldehyde, acetic acid, acrylic acid, acrolein, carbon dioxide, carbon monoxide, nitrogen, oxygen and steam. Based on this, the following reactions can be said to have occurred in the reactor;



8.1.1 Extent of Reactions

Based on the known outlet and inlet, to determine the actual reactions occurring in the reactor, for all the reactions, the extent of reactions was determined where extent of reactions ζ is given by -

$$\zeta_j = \frac{F_{i,inlet} - F_{i,outlet}}{v_{ij}}$$

where, v_{ij} is the stoichiometric coefficient of species “i” in reaction “j”. For each of the 14 reactions, the extent was calculated using the python code mentioned in Annexure 1.

From the computational calculations, the following extent of reactions were obtained

Table 27: Extent of the reactions

number	reaction	extent of the reaction
1	$\text{IB} + \text{O}_2 \rightarrow \text{MAC} + \text{H}_2\text{O}$	38.7000
2	$\text{IB} + 1.5 \text{O}_2 \rightarrow \text{MAA} + \text{H}_2\text{O}$	1.5191
3	$\text{IB} + 5\text{O}_2 \rightarrow \text{Acetic acid} + \text{CO}_2 + 3\text{H}_2\text{O}$	1.1094
4	$\text{IB} + 3\text{O}_2 \rightarrow \text{Acrylic acid} + 2\text{H}_2\text{O}$	0.0303
5	$\text{IB} + 2.5\text{O}_2 \rightarrow \text{Acrolein} + \text{CO}_2 + 2 \text{H}_2\text{O}$	0.0000
6	$\text{IB} + 3\text{O}_2 \rightarrow \text{Acetaldehyde} + 2\text{CO} + 2\text{H}_2\text{O}$	1.1094
7	$\text{IB} + 6\text{O}_2 \rightarrow 4\text{CO}_2 + 4 \text{H}_2\text{O}$	1.1650
8	$\text{IB} + 4\text{O}_2 \rightarrow 4\text{CO} + 4 \text{H}_2\text{O}$	0.6542
9	$\text{IB} + 2.5 \text{O}_2 \rightarrow \text{Acroelin} + \text{CO}_2 + 2\text{H}_2\text{O}$	0.5194
10	$\text{MAC} + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 3\text{H}_2\text{O}$	0.0000
11	$\text{MAC} + 3\text{O}_2 \rightarrow 4\text{CO} + 3\text{H}_2\text{O}$	0.0000
12	$\text{IB} + \text{O}_2 \rightarrow \text{Acetone} + \text{Formaldehyde}$	0.6115
13	$\text{Formaldehyde} + 0.5\text{O}_2 \rightarrow \text{co}$	0.5718
14	$\text{Formaldehyde} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$	0.0000

So we can ignore reaction numbers 5, 10, 11 and 14 for the reactor design part.

8.1.2 Mass balance across the reactor

To have a accurate mass balance, it is necessary to obtain the flow rates of the products in terms of the inlet flow rates, selectivity and the conversion. So if the inlet flow rate of the limiting reactant (for our case Isobutene) is given by F_{IB0} , for oxygen it is given by F_{O20} , for nitrogen it is given by F_{inerts} and for

steam it is given by F_{H_2O} , the selectivities for methacrolein, side products, carbon monoxide and carbon dioxide are Y_a , Y_b , Y_c and Y_d respectively and the conversion of isobutene is given by X_a , the mass balance can be obtained in the table 28. Here Y_a is 0.82, Y_b is 0.1, Y_c is 0.03 and Y_d is 0.05. Since the reactor is continuous and the rate equations will use the partial pressures and concentrations, this is a necessary step to do.

Table 28: Mass balance across the reactor

Species	initial	At any point in the pbr
MAC	0	$FIB0 * X_a * Y_a$
Side products	0	$FIB0 * X_a * Y_b$
CO	0	$FIB0 * X_a * Y_c * 4$
CO ₂	0	$FIB0 * X_a * Y_d * 4$
IB	$FIB0$	$FIB0 * (1 - X_a)$
O ₂	$FO20$	$FO20 - (FIB0 * X_a * Y_a - FIB0 * 4.75 * x_a * Y_b - FIB0 * X_a * Y_c - FIB0 * X_a * Y_d * 6/4)$
INERTS	$Finerts$	$Finerts$
H ₂ O	$FH2O0$	$FH2O0 + FIB0 * X_a * Y_a + FIB0 * X_a * Y_b + FIB0 * X_a * Y_c * 4 + FIB0 * X_a * Y_d * 4$
Total	$FT0$	$FT0 + FIB0 X_a (6 Y_c + 5.5 Y_d - 3.75 Y_b)$

8.1.3 Process design

The values of the partial pressures and the concentrations to be used in the kinetic rate equations will be substituted as the function of the flow rates, selectivities and conversions as given in table 28. Then the design equation for a packed bed reactor was used to give the following ode which was then solved for in Python (annexure 2).

$$\frac{dW}{dX_a} = \frac{FIB0}{-R_{ib}}$$

Here $-R_{ib}$ is the total consumption of isobutene for all the reactions where it is a reactant.

Solving this ode, the weight of the catalyst is obtained as 21618.89 kg.

8.1.4 Catalyst properties

The particle density of the catalyst is said to be 2400 kg/m^3 and the voidage is 21%. The specific area of the catalyst is $3.4 \text{ m}^2/\text{g}$ and the diameter is 30 mm.

8.1.5 Mechanical design

Using the weight of the catalyst and the bulk density, we obtain the volume of the bed as 11.38 m³. Assuming a fixed height to diameter or **H/D** ratio for the catalyst bed, we can find the Height of the bed that will be required. Assuming H/D to be 8 (patents say the H/D ratio has the range of 3-20), we get the Height of the bed as 9.75 m. Assuming the support space as 15%, the height of the cylinder can be found:

$$H_{bed} \times (1 + 0.15) = 9.75 (1.15) = 11.21 \text{ m.}$$

For the safe design, the design pressure and temperature should be 1.1 and 1.5 time the operating pressure and temperature. Hence the design pressure is 2.42 bar and design temperature will be 934.73 K.

The volumetric flow rate of the inlet is calculated by using ideal gas law and it comes out to be 3.17 m³/s. Hence the space time for the reactor is the bed volume divided by the volumetric feed flow rate which comes to be 4.13 sec.

Reactor Sizing Parameters					
$V_{bed} (m^3)$	$Diameter (m)$	$H_{bed}(m)$	$H_{cylinder}(m)$	$A_{cross}(m^2)$	$V_{cylinder} (m^3)$
11.38	1.22	9.75	11.21	1.17	13.09

8.1.5.1 Materials of construction

Understanding the performance of different materials is one of the most essential parameters when designing a reactor vessel. For safe and reliable chemical operations, materials' long-term behavior in reactor core, ability to resist degradation and corrosion must be taken into consideration. Furthermore, the materials of construction must be economical, meet mechanical requirements at specific process conditions and allow the maintenance and replacement. Depending on the situation, carbon steel and stainless steel are most practical and convenient in terms of both mechanical properties and financial budget. However, bismuth molybdate catalyst and water that is utilized for the reaction might cause severe corrosion for carbon steel material, which is why stainless steel is preferred for the reactor construction. Also, stainless steel is satisfactory for machining, welding, hot and cold working, and it has significant thermal resistance and tensile strength. Numerous types of stainless steel are available, and they are generally categorized into three classes, Ferritic, Austenitic and Martensitic. The uniform structure of Austenitic steel provides remarkable corrosion resistance, which is the main reason why it

has widely been utilized in process industries till now. Therefore, the reactor vessel is constructed by Type 304(18/8) Austenitic stainless steel with minimum chromium and nickel content (stable structure).

BS1501	801B
AISI	304
Composition	0.08% C, 2% Mn, 8 – 11% Ni, 17.5 – 20% Cr,
Thermal Conductivity (W/m°C)	16.2
Melting Point(°C)	1400 – 1455
Density (kg/m ³)	8000
Tensile Strength at 380°C (MPa)	114

8.1.5.2 Tube design

Assuming a tube outer diameter of 31.75, length 12 m and a trial thickness of 2mm, we can calculate the thickness using the thickness for a vessel under pressure. We calculate the ratio of outer diameter to assumed thickness and the ratio of length to the outer diameter to get the value of factor A and factor B to get the allowable pressure value. If allowable pressure is more than design pressure, the thickness will be adequate. The following results are obtained from the calculations -

t _{tube_trial}	2
Do/t	15.88
L/Do	353.19
Factor A	5.00E-05
Factor B	5.50
Allowable P (mPa)	0.46

Hence the allowable pressure is 4.6 bar which is more than our design pressure.

Now to get the number of tubes, we calculate the inner volume of the tube using the formula $V = \pi r_i^2 L$ which comes to be 7.26 mm³. The volume of the catalyst should be equal to the number of tubes into the volume of one tube. Hence we get the number of tubes as 1242.

The cross section area of one tube will be calculated as 604.81 mm². Now using this we can get the velocity in one tube using the formula

$Q = \text{Area into velocity into number of tubes}$

So velocity of gas in one tube will be 4.22 m/s.

Using the formula for baffle diameter for square pitch, we get the baffle diameter as 1.81m, if we assume baffle diameter to be 95% of shell diameter, the shell diameter is 1.91m.

8.1.5.3 Shell design

The shell side will have the heat transfer fluid. The chosen thermic fluid in this case is LiNaKNO₃ salt. It will be passed in the molten form. The melting point is 451 K. The calculations for the shell side are given below

cp_salt	1.65 kJ/kgK		
T_in	401 K	T_rise	252.37
T_out (max)	653.37 K		
velocity	0.80 m/s	range is 0.6-	
density	2700 kg/m ³	1.5 m/s	
flow rate	7.80E-02 m ³ /s	Viscosity (Pas)	
Q_req	6.90 MW	210.63 kg/s	
mcp	347.53 kW/K		
delT	19.85		

8.1.6 Heat transfer calculations

Using Dittus Boelter equation we get the heat transfer coefficient on the shell side as 211.08 W/m²K and on the tube side as 12.26 W/m²K. Using these values we design for the hotspot.

8.1.7 Accessories design

Nozzle design

Nozzles are important to control the inlet and outlet flow of the reactors and optimum diameter of the nozzles can be calculated by Harker equation:

$$D_{opt} = 8.41 * \frac{w^{0.45}}{\rho^{0.31}} = 8.41 * \frac{5273.17^{0.45}}{2.68} = 293 \text{ mm}$$

D_{opt} –optimal diameter (mm),

w – mass flow (kg/min)

ρ – density (kg/m³).

Having calculated the optimal diameter, the thickness of nozzles can also be determined:

$$t_{nozzle} = \frac{P_{design} * D_{opt}}{2 * \sigma * \eta - 0.2 * P_{design}} = \frac{5.5 * 10^6 * 0.293}{2 * 100 * 10^6 * 0.9 - 0.2 * 5.5 * 10^6} = 1 \text{ mm}$$

Nozzle reinforcement

Maximal allowable stress near the nozzle decreases and that is why, reinforcing pads should be welded there with 1.5 or 2 times of the thickness of nozzle diameter [37]. Therefore, assumption could be made in this range:

$$D_{reinforcement} = D_{opt} * 1.5 = 293 * 1.5 = 440 \text{ mm}$$

440 mm reinforcement should be installed for the nozzles.

Vessel

Internals

Manhole

20-inch (0.5 meter) manhole must be designed for the reactors with diameter more than one meter. Also, only one manhole is required for the single bed vessels [37]. Therefore, one 0.5 meter manhole will be used MAL reactor.

Bursting disc

Bursting discs are safety devices designed and manufactured to relieve the system pressure if the pressure of the vessel exceeds the predetermined value. Generally, minor pressure increments

can be compensated by the relief valves, however, if the system pressure overruns the 20% of maximal design pressure, then bursting discs fail and provide full hole opening for flow. Depending on the corrosivity of the fluid, variety of construction material for the bursting discs can be used. The size range of bursting discs is from 3 mm to 1.2 m and diameter of the rupture disk can be assumed as 0.5 meter [37].

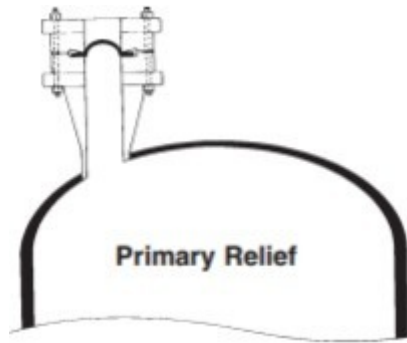


Figure 5. Bursting Disk

Wind Loads and Dynamic Wind Pressure

The load and speed of the wind should be considered when designing equipment for process plants. Therefore, the meteorological factors need to be analyzed elaborately for both onshore and offshore operations. Pressure that the wind impact to cylindrical vessels can be found by the equation below:

$$P_w = 0.05 * u_w^2 \quad (2.8)$$

In which,

P_w – wind pressure (P),

u_w – wind speed (km/hr).

The wind velocity depends on the local topography and some other factors of the chosen location. Hai Pong city of Vietnam is selected as the target location for the MMA plant and maximal velocity of wind in Hai Pong is recorded as 15 km/hr in January which is the windiest month in Hai Pong [40]. However, calculations should be done for 30 km/hr for safer design.

$$P_w = 0.05 * 30^2 = 45 \text{ Pa}$$

Now, the load imposed per reactor length can be evaluated:

$$W = P_w * D_{eff} \quad (2.9)$$

W – load per unit length (N/m),

D_{eff} – effective diameter (m).

Effective diameter is calculated as the sum of reactor outside and insulation diameters, however, for isothermal reactors insulation is not installed and therefore, can be skipped.

$$W = P_w * D_{eff} = 45 * 3.265 = 147 \text{ N/m}$$

Knowing the load per unit length, it is feasible to determine the bending moment:

$$M_x = \frac{(W * H^2)}{2} = 147 * \frac{12.86^2}{2} = 12152.6 \text{ N * m}$$

M_x – maximal bending moment (N*m),

H_{vessel} – overall height of vessel (m).

Weight of Vessel

Approximate weight of the multitubular isothermal steel fixed bed reactor can be calculated by knowing the important parameters [37]:

$$W_v = 240 * C_v * D_m * (H_{vessel} + 0.8 * D_m) * t_{wall}$$

Where,

W_v – reactor weight excluding internals and fittings(kN),

C_v – vessel internals constant accounts for weight of internals (taken as 1.15),

D_m – effective mean diameter, which is the sum of internal diameter and wall thickness (m).

$$D_m = D_{internal} + t_{wall} = 3.24 + 0.014 = 3.25 \text{ m}$$

$$W_v = 240 * 1.15 * 3.25 * (12.86 + 0.8 * 3.25) * 0.014 = 194.5 \text{ kN}$$

8.2 Extraction column

The present report provides a comprehensive design of a liquid–liquid extraction (LLE) system for the recovery of Methacrylic Acid (MAA) from an aqueous feed stream using n-Butyl Acetate (NBA) as the extraction solvent. The feed stream of 6,063 kg/h contains 2,124 kg/h of MAA, and is processed in a counter-current extraction column operating at 13°C. The solvent flow rate is maintained at 10,000 kg/h to ensure favorable mass transfer and high recovery efficiency.

Based on equilibrium analysis and stage-wise calculations, the system requires 5 theoretical stages; however, considering a realistic stage efficiency of 80%, the number of actual stages is determined to be 7. The extraction column selected is a pulsed perforated plate column with a diameter of 0.9 m and a height of 6.5 m. The overall system achieves approximately 95% recovery of MAA.

Process Description

The extraction process is carried out in a counter-current mode, wherein the aqueous feed enters from the top of the column and flows downward, while the organic solvent (NBA) enters from the bottom and rises upward. Due to density differences, the aqueous phase acts as the continuous phase, and the organic phase is dispersed through pulsation.

Mass transfer of MAA occurs at the interface between the two immiscible phases, driven by concentration gradients and governed by equilibrium relationships. The process results in an extract stream rich in MAA and a raffinate stream depleted of MAA.

Thermodynamic Equilibrium

The equilibrium relationship between MAA concentration in the aqueous and organic phases is expressed as:

$$x_{org} = -0.847 x_{aq}^2 + 2.153 x_{aq} + 0.0012$$

$$Y' = 4.234 (X')^{0.892}$$

The distribution coefficient varies with concentration due to molecular interactions such as dimerization, and an average value of 3.5–4.0 is used for design purposes.

Kremser Equation and Stage Analysis

The number of theoretical stages is calculated using the Kremser equation for counter-current extraction:

$$N = \ln \left[\frac{(X'_f - Y'_m)/(X'_r - Y'_m)}{(1 - 1/E) + 1/E} \right] / \ln(E)$$

The extraction factor is given by:

$$E = m' (S'/F')$$

Substituting the design parameters yields an extraction factor of approximately 9.65, and the calculated number of stages assuming constant slope is 1.27. However, due to variation in equilibrium slope, stage-by-stage analysis indicates that 5 theoretical stages are required.

Stage Efficiency and Actual Stages

Considering a stage efficiency of 80%, the number of actual stages is calculated as:

$$N_{\text{real}} = N_{\text{theoretical}} / \text{efficiency} = 5 / 0.80 = 6.25 \approx 7 \text{ stages}$$

6. Column Design

The column diameter is determined based on volumetric flow rate and allowable velocity, resulting in a calculated value of 0.305 m; however, a standard diameter of 0.9 m is selected for safe operation.

The height of the column is calculated using:

$$H = N \times \text{HETS} = 5 \times 0.85 = 4.25 \text{ m}$$

Including disengagement zones and design margin, the total height is taken as 6.5 m.

7. Material Balance

The overall mass balance is satisfied with total input equal to total output. The MAA recovery is calculated as:

$$\text{Recovery} = (2017.8 / 2124.0) \times 100 = 95\%$$

8. Equipment Specifications

The extraction is performed in a pulsed perforated plate column with the following specifications:

Diameter: 0.9 m, Height: 6.5 m, Plates: 85, Plate spacing: 50 mm, Pulsation frequency: 1.5 Hz, Amplitude: 15 mm. The material of construction is SS316L.

9. Conclusion

The designed LLE system successfully achieves 95% recovery of Methacrylic Acid under the specified operating conditions. The design is technically feasible, economically viable, and suitable for industrial implementation.

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